

Daily Progress Report — Day 3

Project Name: MediFlow CRM — Doctor / Clinic Practice Management System

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Reporting Period: Daily

1. Summary of Today's Activities

Today, the team accomplished major development milestones across **Phases 1, 2, 3, and 4** of MediFlow CRM, delivering a fully integrated, production-grade practice management foundation. We set up the application environment and compiled responsive Tailwind CSS v4 assets. We engineered the Authentication & Staff Subaccounts module with 22 configurable permissions across 5 default roles. We implemented the complete 7-tab Practice Settings module with persistent clinic configuration and active session controls. Furthermore, we finalized the Patients Directory and the comprehensive 9-Section Clinical Profile Hub with multi-criteria search, clinical notes logging, and secure medical document uploads. All deliverables have been formatted with Laravel Pint and verified by a **30-test automated PHPUnit suite with 100% pass rate**. We also finalized the architectural plan for **Phase 5 (Appointments & Interactive Calendar)**.

2. Key Discussions & Decisions

A. Architecture, RBAC & Security (Phases 1 & 2)

- **Tailwind v4 & Master Shell Integration:** Established the master application layout with responsive navigation (persistent collapsible desktop sidebar, slide-out mobile drawer, top search/action bar) with Tailwind CSS v4, Alpine.js, and Chart.js integration.
- **Role-Based Access Control (RBAC):** Built the CheckRoleOrPermission middleware enforcing granular access across 22 configurable permissions and 5 system roles (admin_doctor, clinic_manager, receptionist, nurse, accountant). Defaulted sensitive actions (delete, system settings, staff creation) to restricted.
- **Staff Directory & Subaccounts:** Built SubaccountController enabling the Doctor to create, manage, toggle active/inactive status, and configure granular permissions for clinic personnel.

B. Practice Settings Engine (Phase 3)

- **7 Functional Settings Tabs:** Implemented a centralized settings repository storing: Doctor Profile (credentials, registration numbers), Clinic Parameters (working hours, consultation fees, currency), Appointment Rules (15-min slot duration, daily cap of 30, cancellation window), Prescription Defaults, Invoice Numbering (INV-1001), Notification Triggers, and Security Controls (password updates, active session termination).

C. Patients Directory & 9-Section Clinical Profile Hub (Phase 4)

- **18-Field Intake Schema:** Standardized the patient schema to capture 11 demographic fields (Name, DOB, Age, Gender, Blood Group, Phone, Email, Address, Emergency Contact, Relation, Emergency Phone) and 7 clinical fields (Allergies, Chronic Conditions, Current Medications, Past Surgeries, Family Medical History, Smoking, Alcohol).
- **Auto-Generating Medical Identifiers:** Implemented automated sequential PAT-XXXXX patient ID generation.
- **Unified 9-Section Profile Hub:** Developed the unified medical profile screen incorporating all 9 specification sections: Personal Info, Medical History, Appointments, Consultations, Prescriptions, Invoices, Payments, Notes, and Documents/Attachments.
- **Clinical Notes & Secure Document Attachments:** Built clinical note timestamping and authenticated document management with isolated local disk storage (storage/app/patient_documents/) and secure streaming for medical reports (PDF, images, DOCX).
- **Multi-Criteria Search & Filtering:** Implemented search by Patient ID, Name, Phone, and Email, combined with dynamic filters for Gender, Blood Group, Age range, and Registration date.

D. Quality Assurance & Verification

- Created dedicated feature test suites: AuthenticationTest (5 tests), SubaccountTest (9 tests), SettingTest (7 tests), and PatientTest (9 tests).
- Total test suite output: **30 passed, 0 failed, 95 assertions.**
- Formatted all modified PHP classes using Laravel Pint (vendor/bin/pint --format agent).

3. Members Role

Team Member	Assigned Role	Module & Responsibility Breakdown
Anwasha Manna (<i>Backend & Database</i>)	Backend Architecture & Database Lead	• Built migrations and models for settings, patients, patient_notes, and patient_documents • Engineered CheckRoleOrPermission authorization middleware and subaccount permission synchronization • Implemented SettingController repository logic for 7 practice settings tabs and session revocation • Developed PatientController CRUD, auto PAT-XXXXX generation, multi-criteria filtering, and soft-delete archiving • Authored PHPUnit feature tests for Subaccounts, Settings, and Patients

Indranil Dutta (<i>Full-Stack / Clinical Workflow Lead</i>)	Clinical Engine & Workflow Specialist	• Designed and constructed the 9-Section tabbed Patient Profile Hub (resources/views/patients/show.blade.php) • Built the clinical intake form (patients/create.blade.php and edit.blade.php) covering all 18 demographic and medical history fields • Implemented real-time clinical notes recording and authenticated patient document upload/download mechanisms • Formulated the architectural blueprint and lifecycle state transitions for Phase 5 Appointments & Interactive Calendar
Debarati Gantait (<i>Frontend, Billing & Reporting Lead</i>)	Billing, Analytics & UI/UX Lead	• Built and configured the responsive master application shell, desktop sidebar, mobile drawer, and top navigation header • Implemented Tailwind CSS v4 asset compilation pipeline with Vite and Alpine.js component reactivity • Styled the Patient Directory table with touch-friendly responsive mobile card conversion and allergy warning badges • Designed tab navigation and form layouts for Practice Settings and Subaccount permission management • Ran Laravel Pint style formatting and asset re-bundling (npm run build)

4. Roadblocks & Knowledge Gathering Strategy

- **Area of Focus:** Handling multi-field patient filtering across combined demographic and clinical criteria without degrading query performance; ensuring secure document streaming without exposing private storage paths.
- **Action Taken:** Applied indexed Eloquent query scoping with selective whereDate, whereBetween, and normalized substring searches; routed file access through an authenticated controller method using Storage::disk('local')->download() with MIME type and existence verification.
- **Next Focus:** Proceeding to **Phase 5 (Appointments Module & Interactive Calendar)** to implement the 7 lifecycle appointment states (Pending → Confirmed → Checked In → In Consultation → Completed / Cancelled / No Show) and the Month / Week / Day interactive calendar views.

Submitted By GROUP A

(Anwesha Manna, Debarati Gantait, Indranil Dutta)